

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2460883.8077 +/- 0.0077 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.105 +/- 0.042
Transit depth (Rp/Rs)^2	1.1 +/- 0.88 [%]
Orbital Inclination (inc)	86.0 +/- 2.55
Airmass coefficient 1 (a1)	0.987 +/- 0.015
Airmass coefficient 2 (a2)	0.0103 +/- 0.0053
Scatter in the residuals of the lightcurve fit is	1.53 %
Best Comparison Star	#3 - [84, 21]
Optimal Aperture	3.59
Optimal Annulus	11.6
Transit Duration (day)	0.068 +/- 0.019

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.105 ± 0.042 vs 0.1646 (deviation 1.42σ)
Duration fit vs published	0.068 d vs 0.0758 d (deviation 0.41σ)
Mid-time O–C	18.5 min
Depth SNR	2.5
Residual scatter	1.53 %

Light curve

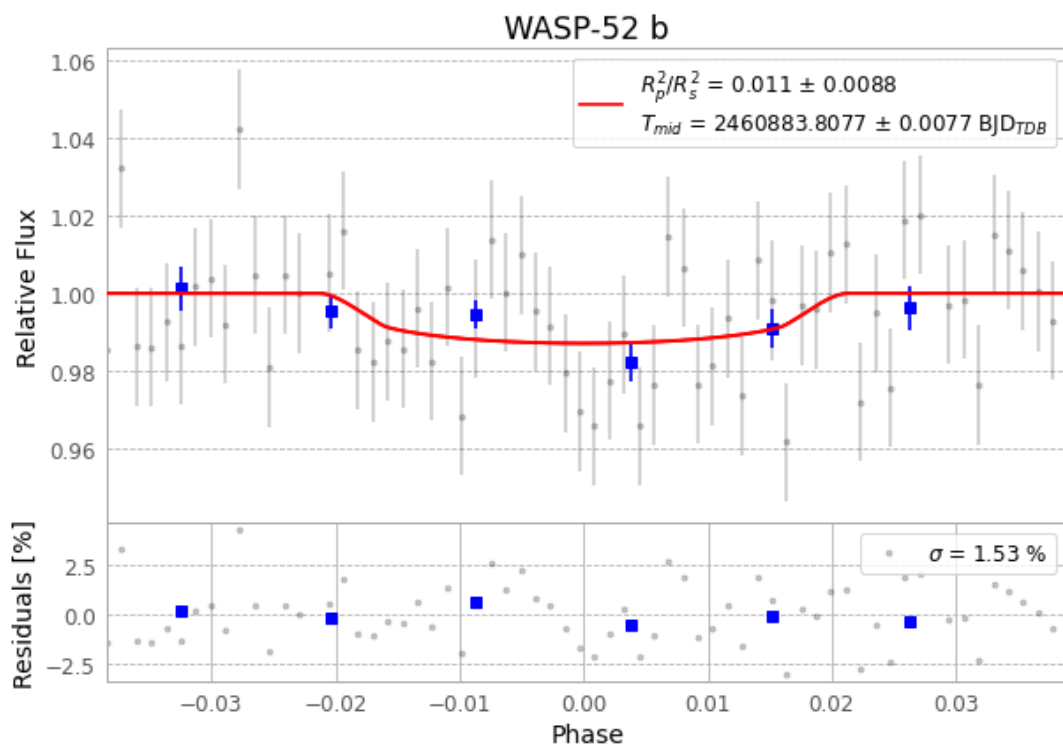


Figure 1 — FinalLightCurve_WASP-52 b_2025-07-26.png (SHA-256 5a174fe877f56d48...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [294, 175], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 d187c0806c10da27...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 39d9b80

Data provenance

66 FITS frames (43 MB), WASP-52250727053916.FITS → WASP-52250727085415.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-52-250727.log (418c73189e48...), FinalLightCurve_WASP-52 b_2025-07-26.png (5a174fe877f5...), FinalLightCurve_WASP-52 b_2025-07-26.csv (6294cfa4139e...)

Compute resources

Wall time 95.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-20 14:51Z.